

• STRATEGIC REVIEW

Allternit Manufacturing

A digital microfactory for AI hardware, B2B production, and distributed manufacturing.

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Bigger than a print farm

Allternit Manufacturing is the physical production arm of the company. It produces our own hardware, monetizes excess capacity, and becomes a rapid prototyping lab — all while building software that can eventually be licensed to other factories.

3

PROBLEMS SOLVED

6

OPERATING DIVISIONS

10

YEAR ROADMAP

Three problems, one division

1

Internal Manufacturing

Print every Allternit product: compute box enclosures, battery housings, rack mounts, cooling ducts, cable management, and development fixtures.

2

Cash Flow

Sell excess machine time to local businesses. Replacement parts, jigs, fixtures, and prototypes — same day, not same month.

3

Rapid Prototyping

Turn weeks of outsourced prototyping into hours. Iterate on hardware designs the same day an idea appears.

The question is no longer *"What should we print?"*

It is **"How do we build a small digital factory that monetizes unused machine time?"**

Six operating divisions



Design Studio

Industrial design, CAD modeling, reverse engineering, 3D scanning, material selection, and DFM/DFA reviews.



Prototype Lab

Same-day iteration. CAD → print → test → revise → print again. Fit checks, airflow tests, layout validation.



Digital Manufacturing

Material-specific print farms: PLA/PETG, ABS/ASA, engineering nylon, carbon fiber, resin, large-format FDM.



Product Manufacturing

Allternit's own catalog: compute enclosures, rack accessories, cable organizers, battery holders, docking stations.



Manufacturing Services

B2B production: replacement parts, jigs, fixtures, low-volume runs, emergency turnaround, engineering consulting.



Hardware R&D

Portable AI stations, edge servers, AI kiosks, robotics platforms, IoT devices, custom electronics enclosures.

Become a bottleneck solver

Don't become "a 3D printing company." Become the company that removes manufacturing bottlenecks for other businesses.

VERTICAL	EXAMPLES WE PRINT	PRICING MODEL
Machine Shops	Drill guides, fixtures, jigs, spacers, alignment tools	Monthly retainer + per-part
HVAC	Duct adapters, mounting brackets, sensor housings	Per-job quoting
Electricians	Junction box adapters, labeling fixtures, conduit tools	Per-job quoting
Construction	Concrete forms, layout templates, measuring tools, pipe spacers	Project-based
Universities	Robotics parts, custom mounts, sensor brackets, experiment hardware	Lab account + grants
Manufacturing Plants	Same-day functional replacements for broken plastic parts	Urgent premium + retainer
Real Estate Developers	Site models, floor-plan 3D models, digital twin geometry, marketing renders	Project-based + licensing

Revenue mix & ladder

Target Mix

SOURCE	TARGET
Internal product manufacturing	25%
B2B industrial printing / MaaS	25%
Engineering & CAD services	20%
CAD-as-a-Service	15%
Software & licensing	5%
Consumer / catalog products	5%
Prototyping for startups	5%

Value Ladder

- › **Level 1** — Print products
- › **Level 2** — Design products
- › **Level 3** — Engineer products
- › **Level 4** — Manufacture products
- › **Level 5** — Automate factories
- › **Level 6** — License factory software

Diversification means we are not dependent on Etsy, a single client, or a single product category.

CAD-as-a-Service & software revenue

Printing is the fulfillment layer. CAD, design, and the software that orchestrates production are the high-margin layers.

CAD-as-a-Service verticals

- › **Real estate developers** — site models, floor-plan 3D models, digital twins, marketing renders
- › **Product companies** — production-ready part files, enclosures, mounting hardware
- › **Startups** — full design packages for crowdfunding/hardware MVPs
- › **Robotics builders** — URDF/SDF assemblies, joint brackets, end-effectors

Software & licensing products

- › **Manufacturing OS** — quoting, scheduling, QA, inventory, customer portal
- › **CAD-as-a-Service API** — text-to-CAD, URDF/SDF generation, DFM checks
- › **Print Farm Management** — fleet monitoring, failure detection, auto-reordering
- › **Network License** — certified partner shops run Allternit software

CAD, 3D AI & digital twin stack

CATEGORY	TOOL	COST	BEST FOR
Parametric CAD	FreeCAD	Free	Mechanical design, enclosures
	Onshape	Paid	Cloud collaboration
	Fusion 360	Paid	Integrated CAD/CAM
3D AI / Generative	text-to-cad	Free	Text → STEP/STL/URDF/gcode
	Blender + MCP	Free	Agent-controlled 3D scenes
	LumaLabs AI	Paid	Photorealistic 3D captures
Digital Twins	PlayCanvas SuperSplat	Free	Browser walkable real estate twins
	Photogrammetry	Free/Paid	3D models from photos

Recommended start: **FreeCAD + CadQuery + text-to-cad** for mechanical automation, **Blender + MCP** for visualization, **SuperSplat** for real estate twins.

Robotics & hardware design resources

Open-source humanoid robots

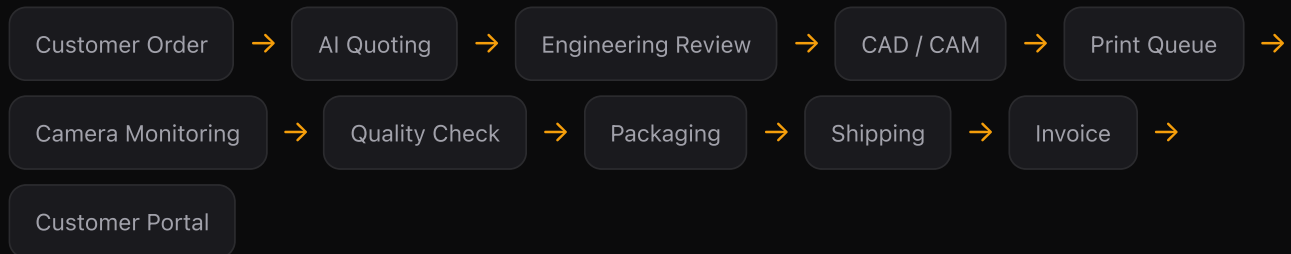
- › **Asimov 1** — ~\$15K DIY kit, full open-source stack
- › **Berkeley Humanoid Lite** — sub-\$5K, 3D-printed
- › **HopeJR** — ~\$3K, Hugging Face
- › **ROBOTIS K0** — full-size open-source platform
- › **Vibe A1** — mini humanoid from \$649

Reference sites & media

- › **humanityslastmachine.com** — interactive humanoid component breakdowns and sourcing
- › **Asimov DIY kit** — <https://asimov.inc/diy-kit>
- › **text-to-cad repo** — <https://github.com/earthtojake/text-to-cad>
- › **Robotics CAD videos** — Desktop/Twitter Bookmarks/organized/tech/robotics-cad/

The Allternit Manufacturing Stack

A software-controlled pipeline where every step can eventually be automated.



Sales Agent

Receives requests, asks clarifying questions, generates quotes.

Engineering Agent

Validates CAD, checks tolerances, suggests DFM improvements.

Manufacturing Agent

Schedules printers, chooses filament, estimates completion.

Operations Agent

Orders filament, tracks inventory, predicts shortages.

Quality Agent

Uses computer vision to detect defects and reject bad parts.

Customer Agent

Updates order status, sends shipping notifications, collects feedback.

Build phases

Phase 1 — Validate

Months 1–3

- 3 printers, 1 workstation, 1 shipping station
- First internal compute box prototype
- 3–5 pilot B2B customers
- \$2k–\$5k monthly revenue

Phase 2 — Scale

Months 4–12

- 10 printers + resin capability
- Standard service packages
- Internal manufacturing queue software
- \$10k+ monthly revenue

Phase 3 — Automate

Year 2+

- 40 printers across material farms
- AI scheduling and failure detection
- Customer portal live
- \$30k+ monthly revenue

• LONG TERM

Ten-year roadmap

YEARS	FOCUS	MILESTONES
1–2	Profitable microfactory	5–20 printers, B2B focus, internal hardware production
3–5	Expanded capabilities	CNC, laser, electronics assembly, proprietary product catalog
5–7	Manufacturing OS	Release Allternit Manufacturing OS: quoting, scheduling, QA, inventory
7–10	Distributed network	Certified partner shops run Allternit software; orchestration layer

The end state: Allternit does not need to own 1,000 printers. It coordinates 1,000 printers through a distributed manufacturing network.

Equipment & materials

Printers

- › **Workhorse:** Bambu Lab P1S / X1 Carbon
- › **Budget starter:** Prusa MK4, Creality K1 Max
- › **Large format:** Creality K1 Max, Voron 2.4
- › **Resin:** Elegoo Saturn 4 Ultra, Formlabs Form 4

Post-Processing

- › Wash + cure station
- › Heat treatment oven
- › Sanding / vapor smoothing station
- › Ultrasonic cleaner

Materials

MATERIAL	USE CASE
PLA	Prototypes, indoor parts
PETG	General purpose, outdoor
ABS / ASA	Heat / UV resistance
Nylon	Wear parts, structural
Carbon-fiber nylon	High stiffness, lightweight
Polycarbonate	Impact resistance
Resin	Detail, dental, miniatures

Fits into everything Allternit builds

Local Brain

Runs CAD, slicing, queue, and vision QA entirely offline.

Computer Use

Monitors printers, captures failure images, interacts with printer UIs.

Swarm ADE

Orchestrates sales, engineering, scheduling, QA, and customer agents.

Allternit Code

Builds and maintains the manufacturing software.

Canvas

Documents procedures, specs, and customer reports.

A://Labs

Trains team and certifies operators on the microfactory workflow.

Catalog ideas

AI Hardware

Compute box shells, battery packs, cooling systems, docking stations, cable organizers, rack accessories.

Networking

Mini rack systems, router mounts, switch brackets, fiber organizers.

Workshop & Office

Laptop stands, phone docks, camera mounts, tool holders, jig systems, drawer organizers.

Emerging

Agriculture irrigation parts, energy battery enclosures, robotics frames, home automation brackets.

Risks & next steps

Key risks

- › Printer downtime → maintain spare parts and redundant capacity
- › Failed prints → camera monitoring + AI failure detection
- › Material moisture → dry cabinets + tracking
- › Tolerance issues → standardize profiles and qualify each printer
- › Regulatory (medical) → stay in industrial/consumer markets first

Immediate next steps

- › Approve master plan and Phase 1 budget
- › Validate CAD-as-a-Service pricing with 2–3 real estate or product clients
- › Select first printers and reserve physical space
- › Build internal part backlog for Allternit products
- › Identify 5 pilot B2B customers for printing or CAD
- › Choose CAD/CAM toolchain (FreeCAD + CadQuery + text-to-cad)
- › Prototype text-to-cad pipeline for one Allternit part

This is a brainstorm turned into a reviewable plan.
Nothing is committed until you decide to move to Phase 1.